ANALYSTLAB AFRICA

DATA ANALYTICS INTERNSHIP

WEEK 5 PROJECT REPORT

SwiftLog Logistics Analytics & AI Transformation Project

Title: Enterprise supply chain Analytics and Artificial Intelligence Transformation for SwiftLog Logistics Limited.

TOOLS: Microsoft Power BI Desktop (Power Querry and DAX)

INTERN: OTUTU LINDA CHINECHEREM

SwiftLog Logistics Analytics & AI Transformation Project

Part 1: Data Import, Understanding & Preparation

1. Project Overview

SwiftLog is a logistics business. This project integrates its operational datasets to carry out a business analysis, develop KPIs and build an interactive Power BI dashboard. This document records Part 1: how the data was imported, inspected and prepared, and how the data model was designed.

2. Data Import

Eleven CSV files were imported into Power BI Desktop using Get Data, then Text/CSV, then Transform Data. Each file was loaded as its own query so it could be inspected and cleaned separately in Power Query. The files were not combined into one table, because they describe different business entities and are linked later through relationships.

Dataset	Description	Columns	Rows
Customers	Customer list with city	CustomerID, CustomerName, City	1,000
Orders	Orders placed by customers	OrderID, CustomerID, OrderDate, Status	5,000
Order_Items	Products and quantities on each order	OrderItemID, OrderID, ProductID, Quantity	15,000
Products	Product catalogue with price	ProductID, Category, UnitPrice	250
Shipments	Shipment for each order with delay in days	ShipmentID, OrderID, WarehouseID, DelayDays	5,000
Delivery_Performance	Delivery status for each shipment	ShipmentID, Status, DelayDays	5,000
Returns	Returns raised against orders	ReturnID, OrderID, Reason	500
Warehouses	Warehouse locations	WarehouseID, Location	15
Employees	Employees assigned to warehouses	EmployeeID, WarehouseID	200
Inventory	Stock held per warehouse and product	WarehouseID, ProductID, Stock	4,000 (3,997 after cleaning)
Suppliers	Supplier list	SupplierID, SupplierName	150

3. Data Inspection

Every dataset was inspected in Power Query using the View tab options Column quality, Column distribution and Column profile, with column profiling set to the entire data set. The checks covered column names, data types, blank values, errors, duplicates and consistency of text values.

3.1 Column names and Data types

Column names were checked against the business meaning of each field and were kept as they appear in the source files, so that key columns have identical names in every table. Data types were verified as follows.

Column(s)	Data type used	Reason
All ID columns (CustomerID, OrderID, ProductID, ShipmentID, WarehouseID, and so on)	Text	Identifiers, not used in arithmetic; text preserves leading zeros such as 000001
OrderDate	Date	Stored in M/D/YYYY format (for example 7/22/2026); orders run from 1 January to 8 September 2026
CustomerName, City, Status, Category, Reason, Location, SupplierName	Text	Descriptive labels
Quantity, DelayDays, Stock	Whole number	Counts and days
UnitPrice	Fixed decimal number	Currency value

3.2 Primary and Foreign keys

Table	Primary key	Foreign key(s)	Notes
Customers	CustomerID	-	Dimension table
Orders	OrderID	CustomerID	One row per order
Order_Items	OrderItemID	OrderID, ProductID	One row per product line
Products	ProductID	SupplierID (assumed, see section 6)	Dimension table
Shipments	ShipmentID	OrderID, WarehouseID	OrderID is also unique, so one shipment per order
Delivery_Performance	ShipmentID	ShipmentID	One row per shipment
Returns	ReturnID	OrderID	An order can have more than one return
Warehouses	WarehouseID	-	Dimension table
Employees	EmployeeID	WarehouseID	
Inventory	None in source (see section 4.2)	WarehouseID, ProductID	The same warehouse and product pair repeats with different stock values

Table	Primary key	Foreign key(s)	Notes
Suppliers	SupplierID	-	No key in the source data links it to other tables

4. Data Quality Assessment and Preparation

4.1 Missing Values

Column quality showed 0% Empty and 0% Error for every column in every table, and no placeholder text such as N/A or null was found. No rows were removed or filled for missing values.

- Inventory contained 3 exact duplicate rows (all three columns identical). They were removed in Power Query by selecting WarehouseID, ProductID and Stock together and choosing Remove Duplicates. The table went from 4,000 to 3,997 rows.
- Selecting all three columns was essential. A warehouse stores many products and a product is held in many warehouses, so repeats in a single column are normal.
- All other tables were checked and contain no duplicate rows. Their key columns (OrderID, CustomerID, ProductID, ShipmentID, ReturnID, and so on) are unique.
- About 1,500 warehouse and product pairs in Inventory appear more than once with different stock values. These are kept as separate stock records and are noted as an assumptions.

4.3 Inconsistent Records

Text columns were checked for stray spaces, mixed capitalisation and spelling variations. None were found. Order Status contains three values (Delivered, Cancelled, Pending), and delivery Status contains two (On Time, Delayed). Trim and Clean were still applied to text columns as good practice (see Part 2).

4.4 Other data observations

- DelayDays appears in both Shipments and Delivery_Performance with identical values.
- In Delivery_Performance, 0 to 2 days of delay is labelled On Time and 3 or more is labelled Delayed.
- 236 orders have no order lines, and 9 customers have never placed an order. Both are kept because they are valid records.
- Many returns belong to orders whose status is Cancelled or Pending. They are kept and noted as a data-quality observation.

4.5 Renaming Fields

The query names were renamed for simplicity and consistency. The columns names were left unchanged to maintain consistency across tables and ensure that the existing DAX measures continues to work perfectly.

5. Data Model

A star-schema style model was built. Transaction tables (Orders, Order_Items, Shipments, Returns, Inventory) are the fact tables. Descriptive tables (Customers, Products, Warehouses, Employees, Suppliers and a date table) are the dimension tables. Every relationship is one-to-many with single-direction filtering, except Orders to Shipments and Shipments to Delivery_Performance, which are one-to-one.

One side (primary key)	Many side (foreign key)	Type
SwiftLog_Customers[CustomerID]	SwiftLog_Orders[CustomerID]	1:*
SwiftLog_Orders[OrderID]	SwiftLog_Order_Items[OrderID]	1:*
SwiftLog_Products[ProductID]	SwiftLog_Order_Items[ProductID]	1:*
SwiftLog_Orders[OrderID]	SwiftLog_Shipments[OrderID]	1:1
SwiftLog_Shipments[ShipmentID]	SwiftLog_Delivery_Performance[ShipmentID]	1:1
SwiftLog_Orders[OrderID]	SwiftLog_Returns[OrderID]	1:*
SwiftLog_Warehouses[WarehouseID]	SwiftLog_Shipments[WarehouseID]	1:*
SwiftLog_Warehouses[WarehouseID]	SwiftLog_Employees[WarehouseID]	1:*
SwiftLog_Warehouses[WarehouseID]	SwiftLog_Inventory[WarehouseID]	1:*
SwiftLog_Products[ProductID]	SwiftLog_Inventory[ProductID]	1:*
SwiftLog_Suppliers[SupplierID]	SwiftLog_Products[SupplierID]	1:* (assumed link)
Dim_Date[Date]	SwiftLog_Orders[OrderDate]	1:* (date table, see Part 2)

6. Data Dictionary: Calculated Fields

These fields were added to the model. The SupplierID column is a DAX calculated column and the rest are DAX measures.

Field	Type	Logic	Purpose
SupplierID	Column (Products)	`"S" & FORMAT(MOD(VALUE(MID(SwiftLog_Products[ProductID], 2, 10)) - 1, 150) + 1, "000")`	Simulated link between products and suppliers (see assumptions)
Total Revenue	Measure	`SUMX(SwiftLog_Order_Items, SwiftLog_Order_Items[Quantity] * RELATED(SwiftLog_Products[UnitPrice]))`	Headline sales value
Net Revenue	Measure	`CALCULATE([Total Revenue], SwiftLog_Orders[Status] <> "Cancelled")`	Revenue excluding cancelled orders
Total Orders	Measure	`DISTINCTCOUNT(SwiftLog_Orders[OrderID])`	Number of orders
Total Customers	Measure	`DISTINCTCOUNT(SwiftLog_Customers[CustomerID])`	Customer base size
Total Products	Measure	`DISTINCTCOUNT(SwiftLog_Products[ProductID])`	Product range size

Field	Type	Logic	Purpose
Total Shipments	Measure	`DISTINCTCOUNT(SwiftLog_Shipments[ShipmentID])`	Delivery volume
Units Ordered	Measure	`SUM(SwiftLog_Order_Items[Quantity])`	Total units demanded
Average Order Value	Measure	`DIVIDE([Total Revenue], DISTINCTCOUNT(SwiftLog_Order_Items[OrderID]))`	Typical value of an order
Average Delivery Delay	Measure	`AVERAGE(SwiftLog_Shipments[DelayDays])`	Average days of delay
Average Delivery Time	Measure	`3 + [Average Delivery Delay]`	Assumed 3-day lead time plus average delay
On-Time Delivery Rate	Measure	`DIVIDE(CALCULATE(COUNTROWS(SwiftLog_Shipments), SwiftLog_Shipments[DelayDays] <= 2), COUNTROWS(SwiftLog_Shipments))`	Share of shipments delayed by 2 days or less
Delayed Shipments	Measure	`CALCULATE(COUNTROWS(SwiftLog_Shipments), SwiftLog_Shipments[DelayDays] > 2)`	Shipments delayed by more than 2 days
Inventory Stock Level	Measure	`SUM(SwiftLog_Inventory[Stock])`	Total units in stock
Inventory Turnover	Measure	`DIVIDE([Units Ordered], [Inventory Stock Level])`	How fast demand uses stock
Warehouse Utilization	Measure	`DIVIDE([Inventory Stock Level], 100000 * DISTINCTCOUNT( SwiftLog_Warehouses[WarehouseID]))`	Stock held compared with assumed capacity
Low Stock Items	Measure	`CALCULATE(COUNTROWS(SwiftLog_Inventory), SwiftLog_Inventory[Stock] < 100)`	Stock records below 100 units
Total Returns	Measure	`COUNTROWS(SwiftLog_Returns)`	Number of returns
Return Rate	Measure	`DIVIDE(DISTINCTCOUNT(SwiftLog_Returns[OrderID]), [Total Orders])`	Share of orders with a return
Cancellation Rate	Measure	`DIVIDE(CALCULATE([Total Orders], SwiftLog_Orders[Status] = "Cancelled"), [Total Orders])`	Share of orders cancelled
Active Customers	Measure	`DISTINCTCOUNT(SwiftLog_Orders[CustomerID])`	Customers who ordered
Repeat Customer Rate	Measure	`VAR Repeaters = COUNTROWS(FILTER( VALUES(SwiftLog_Orders[CustomerID]), CALCULATE(COUNTROWS(SwiftLog_Orders)) > 1)) RETURN DIVIDE(Repeaters, [Active Customers])`	Customer retention indicator
Avg Orders per Customer	Measure	`DIVIDE([Total Orders], [Active Customers])`	Buying frequency
Supplier Performance Score	Measure	`100 * (0.5 * OnTimeRate + 0.3 * (1 - ReturnRate) + 0.2 * (1 - CancellationRate)) (each rate evaluated through the order lines of the supplier with CROSSFILTER)`	Score from 0 to 100 per supplier

7. Assumptions and Limitations

- Supplier link: the source data has no key connecting suppliers to products. Each product was assigned a SupplierID by cycling through S001 to S150 in product order. This is a simulated mapping used to demonstrate supplier analysis. It does not reflect real supplier relationships.
- Delivery time: the data has no dispatch or delivery dates, only DelayDays. Average Delivery Time therefore uses an assumed standard lead time of 3 days plus the average delay.
- On-time delivery: a shipment is on time when DelayDays is 2 or less. This matches the labels in the Delivery_Performance table.
- Warehouse capacity: no capacity data exists. Utilization assumes 100,000 units per warehouse.
- Low stock: stock records below 100 units are treated as low stock.
- Revenue: order lines hold quantities but no prices, so revenue is Quantity multiplied by the current UnitPrice from Products. Total Revenue includes cancelled orders and Net Revenue excludes them.
- Inventory turnover: calculated as units ordered divided by total stock. No cost-of-goods or average-inventory data is available.
- Returns: returns are recorded per order, not per product. Any product-level return figure counts the items in returned orders.
- Retention indicator: the share of ordering customers who placed more than one order.
- Supplier score weights: 50% on-time delivery, 30% returns and 20% cancellations. These weights are a judgement.
- Repeated Inventory pairs and data-quality observations (section 4) were kept rather than deleted.

8. Conclusion

All 11 datasets were imported, inspected and checked for missing values, duplicates and inconsistencies. The only duplicates, 3 exact rows in Inventory, were removed. Primary and foreign keys were identified, relationships were built into a star-schema style model, and every calculated field and assumption was documented. The prepared model supports the transformations in Part 2 and the KPIs and dashboard in Parts 3 and 4.